WORKED SOLUTION

STEP I 2016, Question 4

Pure • Curvature of Curves

STEP I 2016, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q4

The key idea

The expression for κ looks like a hard second-order differential equation, but part (i) is there to show that it is an exact derivative. Writing $p = \mathrm{f}'(x)$, the curvature is the x-derivative of $p/\sqrt{1+p^2}$, so a constant curvature can be integrated once at sight. A second integration is then routine, and the answer turns out to be a familiar shape.

Solution

Part (i)

Using the product rule on $z(1+z^2)^{-1/2}$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{z}{(1+z^2)^{1/2}}\right) &= (1+z^2)^{-1/2} - z\cdot z(1+z^2)^{-3/2} \\ &= \frac{(1+z^2)-z^2}{(1+z^2)^{3/2}} = \frac{1}{(1+z^2)^{3/2}}.\end{aligned}$$

Part (ii)

Let κ be a non-zero constant and write $p = \mathrm{f}'(x)$, so $\mathrm{f}''(x) = \dfrac{\mathrm{d}p}{\mathrm{d}x}$. By part (i) and the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{p}{\sqrt{1+p^2}}\right) = \frac{1}{(1+p^2)^{3/2}}\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{\mathrm{f}''(x)}{(1+(\mathrm{f}'(x))^2)^{3/2}} = \kappa.$$

Integrating with respect to x, there is a constant, which we write as $-\kappa a$ (possible since $\kappa \neq 0$), with

$$\frac{p}{\sqrt{1+p^2}} = \kappa(x-a). \qquad (*)$$

Ask yourself How do we solve $(*)$ for p without losing track of signs?

Squaring is safe if we remember what it throws away. Since $\sqrt{1+p^2} > 0$, $(*)$ says that p has the same sign as $\kappa(x-a)$, and it also says $|\kappa(x-a)| < 1$, because $|p| < \sqrt{1+p^2}$. Squaring, $p^2 = \kappa^2(x-a)^2(1+p^2)$, so

$$p^2 = \frac{\kappa^2(x-a)^2}{1-\kappa^2(x-a)^2}, \qquad \text{and, choosing the sign as above,} \qquad p = \frac{\kappa(x-a)}{\sqrt{1-\kappa^2(x-a)^2}}.$$

Conversely, this p does satisfy $(*)$, since $1+p^2 = 1/(1-\kappa^2(x-a)^2)$.

Now integrate again. Since $\dfrac{\mathrm{d}}{\mathrm{d}x}\sqrt{1-\kappa^2(x-a)^2} = \dfrac{-\kappa^2(x-a)}{\sqrt{1-\kappa^2(x-a)^2}}$, we get, for some constant b,

$$y = b - \frac{1}{\kappa}\sqrt{1-\kappa^2(x-a)^2}, \qquad |x-a| < \frac{1}{|\kappa|}.$$

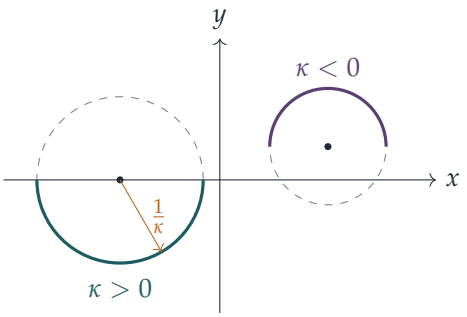
Rearranging, $\kappa(y-b) = -\sqrt{1-\kappa^2(x-a)^2}$, and squaring gives

$$(x-a)^2 + (y-b)^2 = \frac{1}{\kappa^2}.$$

So every curve of constant non-zero signed curvature κ is part of a circle of radius $1/|\kappa|$, centre (a,b), and every such arc arises: reversing the steps, each function above has signed curvature exactly κ. More precisely, $\kappa(y-b) \le 0$, so:

- if $\kappa > 0$, the curve is the lower half $y = b - \sqrt{\kappa^{-2} - (x-a)^2}$, lying below its centre and bending upwards ($f'' > 0$);
- if $\kappa < 0$, the curve is the upper half $y = b + \sqrt{\kappa^{-2} - (x-a)^2}$, lying above its centre and bending downwards ($f'' < 0$).

The endpoints $x = a \pm 1/|\kappa|$ are excluded, since there the gradient is infinite; a curve $y = f(x)$ can only be a graph, so it can never be more than an open semicircle (or a piece of one).



Geometrical significance

For these curves $|\kappa|$ is the reciprocal of the radius of the circle, so a larger $|\kappa|$ means a smaller, more tightly bending circle. The sign of κ records which way the curve bends:

$\kappa > 0$ when the centre of the circle lies above the curve (the curve is concave up), and $\kappa < 0$ when the centre lies below it.

Checks

- Take $\kappa = \frac{1}{2}$, $a = b = 0$: the curve is $y = -\sqrt{4 - x^2}$, the lower half of the circle of radius 2. At $x = 1$, $y = -\sqrt{3}$, $y' = x/\sqrt{4 - x^2} = 1/\sqrt{3}$ and $y'' = 4/(4 - x^2)^{3/2} = 4/3^{3/2}$, so $1 + y'^2 = \frac{4}{3}$ and $\kappa = \dfrac{4/3^{3/2}}{(4/3)^{3/2}} = \frac{1}{2}$, as it should be.
- A straight line has $\mathrm{f}'' = 0$, so $\kappa = 0$: the limit of a circle as the radius $1/|\kappa| \to \infty$, which is why the question excludes $\kappa = 0$.
- Solving $\mathrm{f}'' = \kappa(1 + \mathrm{f}'^2)^{3/2}$ step by step by computer from random starting values, the points stayed on the predicted circle to six decimal places.

Common slips

- Forgetting the constant of integration at the first stage, which loses the freedom to move the centre horizontally.
- Taking the wrong square root for p, which gives the semicircle on the wrong side of the centre and the wrong sign of κ.
- Saying only "κ is the radius": it is the reciprocal of the radius, with a sign showing the direction of bending.

Beyond STEP

Curvature is the starting point of the differential geometry of curves. For a curve parametrised by arc length s, $\kappa = \mathrm{d}\psi/\mathrm{d}s$, where ψ is the angle the tangent makes with the x-axis; here $p/\sqrt{1 + p^2} = \sin\psi$, so $(*)$ is really the statement that ψ changes at a constant rate along the curve. In three dimensions curvature is joined by torsion, and the Frenet–Serret formulae show that these two functions determine a curve up to rigid motion. Pressley's *Elementary Differential Geometry* is a friendly first-year-level introduction.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.